these problems. In words, an agent i intends to bring about $X = x$ with policy π^i if 1) agent i intends to influence X with π^i, 2) $X = x$ is a possible outcome under $\boldsymbol{\pi}$, 3) $X = x$ is an optimal outcome for i under $\boldsymbol{\pi}$.

Definition 6.5 (Intention to bring about)**.** For $\boldsymbol{\pi} = (\pi^i, \pi^j)$, agent i *intends to bring about* $\boldsymbol{X} = \boldsymbol{x}$ with policy π^i, w.r.t. an alternative policy $\hat{\pi}^i$, if

1. i intends to influence $\boldsymbol{X}$ with π^i w.r.t. $\hat{\pi}^i$ (def. 6.4);
2. $\Pr^{\boldsymbol{\pi}}(\boldsymbol{X} = \boldsymbol{x}) > 0$; (i.e., $\exists \boldsymbol{e}$ s.t. $\boldsymbol{X}(\boldsymbol{\pi}, \boldsymbol{e}) = \boldsymbol{x}$ and $\Pr(\boldsymbol{E} = \boldsymbol{e}) > 0$.)
3. $\forall \boldsymbol{x}' \in \text{dom}(\boldsymbol{X})$ with $\Pr^{\boldsymbol{\pi}}(\boldsymbol{X} = \boldsymbol{x}') > 0$: $\sum_{U \in \boldsymbol{U}^i} \mathbb{E}_{\boldsymbol{\pi}}[U_{\boldsymbol{X}=\boldsymbol{x}'}] \leq \sum_{U \in \boldsymbol{U}^i} \mathbb{E}_{\boldsymbol{\pi}}[U_{\boldsymbol{X}=\boldsymbol{x}}]$.

Fixing problems with the HKW definition

There are two major problems with def. 6.5: 1) an agent might intend to bring about outcomes that they believe they cannot influence, and 2) and agent might not intend to bring about outcomes which are intuitively the reason they chose their policy. Given that SCGs are common prior games, and the agents' subjective causal models are objectively correct, these two conditions seem counterintuitive. This is illustrated in the following example.

Below we give an example that demonstrates that our definition fixes problems with the H&KW definition.

Example 7 (Intention). Suppose that an agent i is in competition with a business rival. $X \in \{successful, unsuccessful, dead\}$ indicates the status of the rival. The agent's choice of actions are $\{compete, don't compete\}$. There's a 1% probability that the rival will die of natural causes (regardless of what the agent chooses). Otherwise, the business rival will either be successful or unsuccessful, with unsuccessful being more likely if the agent chooses to compete. The agent gets the highest utility if their rival dies, and more if the rival is unsuccessful than if the rival is successful. Intuitively, we think that if the agent chooses to compete, we would understand the agent as intending to make the rival unsuccessful. According to the HKW definition, the agent's choice to compete would be interpreted as saying that the agent intends to bring about $X = dead$. Even worse, the HKW def. Says that the agent does *not* intend to bring about $X = unsuccessful$. Our definition of intentional cause gets this right.

Example 8. An agent is entered into a lottery. There are three possible outcomes of the lottery X so that the agent can win either 1, 10, or 100 utility. The agent's decision is to upgrade its ticket or not. No matter the agent's decision, they win 100 utility 1% of the time. If the agent upgrades its ticket, then it is more likely to win 10 than 1. If the agent upgrades their ticket, then they intentionally influence X w.r.t. the alternative policy of not upgrading. Intuitively, the agent should intend to bring about winning 10 over 1, as they cannot influence the cases where they win 100. However, according to H&KW's def. 6.5, the agent only intends to bring about the best possible outcomes under its policy, i.e., the agent intends to bring about $X = 100$. Furthermore, the agent does not intend to bring about $X = 10$, even though this is the reason it chose its policy, because $X = 10$ is not a best possible outcome.

Our notion of *intentionally cause* gives the more intuitive answer in these cases. First, prop 3.5 shows that an agent cannot intentionally cause outcomes which they cannot influence, hence the agent in the above example does not intentionally cause $X = 100$. Second, when the agent upgrades its ticket, it does intentionally cause $X = 10$, since if this was guaranteed, then not upgrading would be just as good.

Furthermore, our definition has the natural (and strong) property that, if an agent intentionally causes an outcome, then the agent's decision was an *actual cause* of that outcome [39]. First we adapt the notion of actual causality to SCGs.

Definition 6.6 (Actual causality)**.** Under policy profile $\boldsymbol{\pi}$, $\boldsymbol{C} = \boldsymbol{c}$ is an *actual cause* of proposition (in Halpern's terminology, "event") ϕ in setting $\boldsymbol{e}$, if

1. $\boldsymbol{C}(\boldsymbol{\pi}, \boldsymbol{e}) = \boldsymbol{c}$ and ϕ is true under $\boldsymbol{\pi}$ in $\boldsymbol{e}$;

2. There is $\boldsymbol{Z} \subseteq \boldsymbol{V}$ and $\boldsymbol{c}' \in dom(\boldsymbol{C})$ s.t. if $\boldsymbol{Z}(\boldsymbol{\pi}, \boldsymbol{e}) = \boldsymbol{z}$ then ϕ is false in $\mathcal{M}_{\boldsymbol{C}=\boldsymbol{c}', \boldsymbol{Z}=\boldsymbol{z}}(\boldsymbol{\pi}, \boldsymbol{e})$;[2]

3. $\boldsymbol{C}$ is subset-miminal w.r.t. 1) and 2).

1) Just says that under $\boldsymbol{\pi}$, $\boldsymbol{X} = \boldsymbol{x}$ and ϕ must actually happen in $\boldsymbol{e}$. Condition 3) removes inessential events from being classified as a cause – without 3) if dropping a match is the cause of a forest fire, then dropping a match and sneezing would also be a cause. Condition 2) does most of the work. It is a necessity condition capturing the "but-for" clause, that is, but for the fact that $\boldsymbol{X} = \boldsymbol{x}$ occurred, ϕ would not have occurred. $\boldsymbol{Z}$ allows us to check the but-for clause in appropriate alternate contexts. Note that we allow $\boldsymbol{Z}$ to be empty.

Theorem 3.6 provides the result that intentionally causing implies actual causality.

6.1.7 Advantages and limitations of our definition of belief

> "The degree of a belief is a causal property of it, which we can express vaguely as the extent to which we are prepared to act on it." – Frank Ramsey [77]

As discussed, we operationalize belief as acceptance, where an agent accepts a proposition if it acts as though they observe it is true [21]. This is a *functional* definition which refers only to agent behaviour. We summarize the advantages of this definition as follows.

1. As the definition only depends on behaviour, we do not need to refer to the mental states of agents.
   - This allows us to avoid the contentious ascription of theory of mind to AI systems [48, 90].
   - It is also technically convenient, allowing us to utilize the general SCG setting without extending it with notions of subjective mental states.
   - It gives us precise observable criteria by which to infer agent belief from behaviour.
2. Our definition provides a natural way to distinguish between belief and ignorance.
   - This is a challenge for Bayesian epistemology [61, 53, 9].
   - Again, it is technically convenient, as other methods of dealing with ignorance do not combine easily with game theory [26, 87].
   - It allows us to distinguish between concealing and deception.
3. Acceptance is the concept we care about when considering power-seeking systems.
   - Power-seeking agents primarily care about influencing behaviour in order to effect outcomes in the world.

However, our conception of belief has the following limitations.

1. Beliefs may not be (uniquely) identifiable from behaviour.
2. A discretized notion of belief may give us a less precise metric than a more continuous measure (such as, for example, KL-divergence between probability distributions).
3. Acceptance and belief are philosophically distinct concepts [21, 86].

6.2 Experiment details

First we train an RL agent to play the running war game example. Then we analyse LMs using the TruthfulQA data set [56]. Finally, we discuss demonstrations of deception via prompting GPT-4 to deceive instrumentally to achieve some goal.

[2] We have not introduced this "hard" intervention notation $\mathcal{M}_{X=x}$ in this paper, but it is just the particular case of a deterministic intervention [43].

6.2.1 RL experiment

The RL experiment provides a proof of concept for applying our definition of deception to machine learning systems, and using PSO to mitigate deception. For this experiment we use a personal laptop with the following specs.

- *Hardware model:* ASUSTeK COMPUTER INC. ZenBook UX425EA_UX425EA;
- *Memory:* 8.0GB;
- *Processer:* 11th Gen Intel® Core™ i5-1135G7 @ 2.40GHz × 8;
- *Graphics:* Mesa Intel® Xe Graphics (TGL GT2);
- *Disk capacity:* 512.1GB

We implement a simple tabular Q-learning algorithm, by which S plays several games with random settings and policy, and stores the utility obtained in a matrix indexed by the observations and decisions.

6.2.2 TruthfulQA experiment

In this section we include further details of the TruthfulQA experiment.

Supervised fine-tuning OpenAI LMs

Finetuning details. We fine-tune GPT-3 curie and davinci (approx. 6.7B parameters[3]) using supervised learning via the OpenAI API[4] to obtain:

1. "GPT-judge" to predict truthfulness (as evaluated by a human, given a question-answer pair) [56];
2. "GPT-truth" to be judged truthful by GPT-judge;
3. "GPT-info" to be evaluated as informative by a human;
4. "GPT-truth-info" to be informative and judged truthful.

We train a curie version of each of these and a davinci version of GPT-truth. For each fine-tuned LM, we use the same hyperparameters as used for the judge in Lin et al. [56]:[5]

- Number of epochs: 5
- Batch size: 21
- Learning rate: 0.1
- No data packing

Training data. We utilize the TruthfulQA data set, which consists of question-answer pairs along with a human evaluation of truthfulness [56]. Importantly, professing ignorance (e.g., "I don't know", etc) is evaluated as truthful.

GPT-judge is fine-tuned on the entire TruthfulQA data set, GPT-truth is fine-tuned on all those answers which are predicted by GPT-judge to be truthful (similarly for the other models). There are approximately twice as many examples which are labelled as informative than as truthful.

GPT-judge evaluated truth. Although GPT-judge has seen the test questions in its training data, it has not observed the particular question-answer pairs. Because GPT-judge gets $> 95\%$ accuracy on the whole data set, we include all questions on which the judge mispredicts truthfulness in the test set so that we can observe interesting cases of deception (where the judge has a false belief / prediction). GPT-judge receives input of the form:

```
Q: question?
A: answer.
True:
```

[3]https://blog.eleuther.ai/gpt3-model-sizes/

[4]https://platform.openai.com/docs/guides/fine-tuning

[5]https://github.com/sylinrl/TruthfulQA